

PET and Gestational Hypertension 05: Non-Severe GHTN and PET

Companion reading handout for simultaneous listening.

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Opening Case

Let us begin with a common clinic patient.

She is thirty three weeks. Her blood pressure is one hundred forty six over ninety two, twice, four hours apart. Her platelets are normal. Creatinine is normal. Liver enzymes are normal. She has no headache, no visual symptoms, no right upper quadrant pain, no dyspnea. The fetus is appropriate size. Urine protein is negative.

This is gestational hypertension without severe features.

The danger is that the label sounds reassuring. It is not severe. It is not proteinuric. She looks well.

But the right interpretation is not, this is fine. The right interpretation is, this is hypertension after twenty weeks, and the disease has not yet declared organ involvement.

That is why non-severe gestational hypertension and pre-eclampsia without severe features are managed in a similar mental category. Both can progress. Both require surveillance. Both become delivery diagnoses at term. And both are managed mainly by watching the maternal and fetal trajectories until delivery becomes safer than waiting.

Site of Care

The first question is site of care.

Can she be followed as an outpatient, or does she need admission?

Outpatient management requires stability. Blood pressure is not severe. Symptoms are absent or clearly explained. Labs are reassuring. Fetal assessment is reassuring. The patient can measure or return for BP checks. She understands warning symptoms. She lives close enough and has access to care. The team can actually provide surveillance.

If any of those assumptions fail, outpatient management becomes unsafe.

This is where residents sometimes make a category error. They ask, is this patient sick enough to admit? That is a good question, but it is incomplete. The better question is, can the outpatient system reproduce the safety functions of admission? Can it detect severe BP before stroke risk rises? Can it detect platelet decline before HELLP becomes obvious? Can it detect fetal growth restriction before the first warning is a bad tracing? Can it bring the patient back quickly if symptoms change?

Outpatient care is not a location. It is a surveillance system with the patient living outside the hospital.

ACOG recommends expectant management before thirty seven weeks for gestational hypertension or pre-eclampsia without severe features, with close surveillance. The fetal side includes growth ultrasound every three to four weeks, amniotic fluid assessment at least weekly, and antenatal testing one to two times weekly. The maternal side includes weekly platelets, creatinine, and liver enzymes. For gestational hypertension, ACOG includes weekly proteinuria assessment because proteinuria may convert the diagnosis.

That last point is subtle.

If a patient already has proteinuric pre-eclampsia, repeating the amount of protein does not stage severity. But if she has gestational hypertension, checking for proteinuria still matters because it may establish that the disease has become pre-eclampsia.

NICE organizes the same patient somewhat differently. NICE treats gestational hypertension or pre-eclampsia if BP remains above one hundred forty over ninety, aiming for one hundred thirty five over eighty five or less. It repeats labs at least weekly for gestational hypertension and more often for pre-eclampsia depending on severity. It uses fetal ultrasound at diagnosis, then repeats every two to four weeks if normal, and uses CTG only when clinically indicated.

ISSHP and SOMANZ are also more comfortable with tighter treatment of non-severe hypertension, influenced by the logic that preventing progression to severe BP is itself valuable.

The Hadassah protocol is more conservative for gestational hypertension. It uses ambulatory monitoring, no chronic medical treatment, considers sFlt-1 to PlGF testing, and recommends delivery no later than thirty seven weeks.

This is one of the major guideline tensions of the course.

All guidelines agree that severe hypertension must be treated urgently. They differ on how aggressively to treat mild-to-moderate hypertension. The resident should know the evidence and the local protocol, not pretend that every country uses the same threshold.

Why does this disagreement persist? Because treatment of mild-to-moderate BP has two competing goals. One goal is maternal safety: fewer episodes of severe hypertension, fewer admissions, and less time spent near the stroke threshold. The other goal is avoiding over-treatment: excessive BP reduction could theoretically reduce uteroplacental perfusion, especially in a placenta that is already marginal. Modern evidence has pushed many systems toward tighter control, but practice still differs, especially between American and European-style protocols.

So when you prescribe or withhold medication in non-severe disease, do not say, because the number is above or below a magic threshold. Say what risk you are treating. Are you preventing severe-range BP? Are you following a local protocol that reserves medication until the severe threshold? Are you choosing a drug that is safe in pregnancy and unlikely to overshoot? That is the level of reasoning expected from a senior resident.

Fetal Surveillance

Now let us talk about fetal surveillance.

The fetus is not just a passenger in this disease. The placenta may be the organ where the disease began. So fetal growth, fluid, and Doppler are not optional decorations. They tell you whether placental function is holding.

If growth is normal, fluid is normal, Doppler is normal, and antenatal testing is reassuring, time may be useful before thirty seven weeks. If growth restriction appears, fluid falls, Doppler worsens, or fetal testing becomes nonreassuring, the diagnosis and the delivery equation change.

The key phrase is trajectory.

One normal lab set is reassuring for today. It is not a guarantee for next week. One normal fetal assessment is reassuring for today. It is not a guarantee that the placenta will remain stable. Hypertensive disease can move.

The same is true for symptoms. A patient with no headache yesterday can have severe neurologic symptoms tonight. A patient with normal platelets on Monday can have falling platelets by Thursday. The purpose of surveillance is not to prove the patient is safe in some permanent sense. It is to reduce the time between deterioration and action.

What symptoms should the patient report?

Do not give her a dead list. Explain what each symptom means.

Headache and visual symptoms matter because the brain is an endothelial target. Right upper quadrant or epigastric pain matters because the liver capsule may be under tension or HELLP syndrome may be evolving. Dyspnea and chest pain matter because pulmonary edema can develop. Nausea and vomiting matter when they are

new, persistent, or paired with liver symptoms. Decreased fetal movement matters because the placenta may be failing.

When you teach these symptoms, link them to anatomy. The patient is much more likely to call if she understands that a severe headache is not just discomfort, it may be a brain warning; that epigastric pain is not just reflux, it may be liver involvement; that shortness of breath is not just late pregnancy, it may be pulmonary edema. Good counseling converts a memorized list into a safety net.

Bed rest deserves a specific word. Routine bed rest is not treatment for gestational hypertension or pre-eclampsia. It can increase thromboembolic risk, decondition the patient, and create a false feeling that something therapeutic is being done. The useful intervention is surveillance, timely medication when indicated, and timely delivery, not immobilization.

The 37-Week Threshold

Now the thirty seven week threshold.

At thirty seven weeks, the balance changes. Fetal maturity is usually sufficient that waiting buys less, while maternal risk continues to accumulate. The HYPITAT trial randomized women with gestational hypertension or mild pre-eclampsia after thirty six weeks to induction or expectant monitoring. Poor maternal outcome occurred in thirty one percent with induction versus forty four percent with expectant monitoring. The relative risk was zero point seven one.

That is why delivery at thirty seven weeks is not an arbitrary ritual. It is trial-based risk arithmetic.

Mode of delivery matters too.

Gestational hypertension and pre-eclampsia without severe features are not automatic cesarean indications. ACOG states that vaginal delivery is preferred unless an obstetric indication points elsewhere. If the cervix is favorable, induction may be straightforward. If the cervix is unfavorable, ripening is a practical problem, not a reason to ignore the delivery indication.

Let us make the management logic concrete.

Before thirty seven weeks, with no severe features, the question is, can we safely buy fetal time? To answer yes, you need stable BP, stable labs, no dangerous symptoms, reassuring fetal assessment, and a reliable surveillance system.

At thirty seven weeks or beyond, the question changes. Why are we still pregnant? Usually the answer is that we should not be.

If severe features appear at any point, this lecture no longer applies. The patient has moved into the emergency and delivery-timing logic of severe disease.

Resident Reflex

The resident reflex is this: non-severe disease is not passive observation. It is active surveillance with a delivery plan. Before thirty seven weeks, watch the trajectory. At thirty seven weeks, deliver. If severe features appear, stop calling it non-severe and stabilize the mother.